Derivatives Review Problems

Instructions: In this Problem Set, you'll carefully solve one of these differentiation problems, then make a purposeful mistake somewhere. Follow these steps:

(a) Choose one of the problems below.

(b) Refer to the list on the next page to identify **all** the derivative rules you're going to need to use for your problem.

(Hint: I have chosen all the problems such that the chain rule is definitely required.)

(c) Carefully find the derivative **step by step**, write out **every step**, and list all the rules you used in each step. See the example on the next page to see how detailed your steps should be.

(d) Choose a **second** problem, and do the same things, but make your answer wrong somehow. Bonus points[1] for an error you think people probably make in real life.

Throughout, carefully label any derivative you find by name, using correct derivative notation.

Here are the problems you can choose from:

(1) Find $\dfrac{dy}{dx}$ if $y = (2x-1)^8 - x + e$.

(2) If $L(y) = (\cos(y))^3 + \cos(3)$, find $L'(y)$.

(3) Find $B'(x)$ if $B(x) = \cos(e^{kx})$, where k is a constant.

(4) Differentiate $W(x) = \sqrt{\dfrac{x^4 - 6x + 10}{7x+8}}$.

(5) Differentiate $g(q) = \sin(\sin(\sin(q)))$.

(6) Differentiate $B(x) = \left(7x^5 - 2x^3\right)^4$.

(7) Differentiate $f(y) = \left(\sqrt{y^4+1}\right)\left(\ln\left(\dfrac{y}{2}\right)\right)$.

(8) Find $\dfrac{dB}{dx}$ if $B = \dfrac{e^{x^2}}{\sin(\pi x)}$.

(9) Differentiate $h(p) = \dfrac{(2p^3-1)^{1/4}}{\cos(p+5)}$.

(10) Find $P'(x)$ if
$P(x) = \dfrac{1}{\sqrt{2\pi}\sigma} \cdot \exp\left(\dfrac{-(x-\mu)^2}{2\sigma^2}\right)$.
μ and σ are constants and $\exp(\ldots)$ is another word for $e^{(\ldots)}$

(11) Differentiate $V = \left(3x - \frac{1}{4}\right)^{-4} - (5x)^{5/4}$.

(12) Differentiate $h(x) = \sqrt{\ln(\cos(x))}$.

(13) Differentiate $h(t) = -3(2t^4+7)^4 + 3$.

[1] Points don't exist.

(Example) Find $\dfrac{dp}{dt}$ if $p(t) = \dfrac{\cos(t^2-1)}{e^{-3t}}$.

I am going to need the following derivative rules:

- quotient rule
- power rule
- exponential function rule
- trig function rule (sine)
- sum rule
- constant multiple rule
- chain rule
- constant function rule

Now I will find the derivative, step by step, and explain what rules I am using in every step. I am highlighting changes from the previous line in blue and with a box around it.

Steps	**Rules**
$\dfrac{dp}{dt} = \dfrac{[e^{-3t}]\cdot\dfrac{d}{dt}[\cos(t^2-1)] - [\cos(t^2-1)]\cdot\dfrac{d}{dt}[e^{-3t}]}{[e^{-3t}]^2}$	(quotient rule)
$= \dfrac{[e^{-3t}]\cdot\dfrac{d}{dt}[\cos(t^2-1)] - [\cos(t^2-1)]\cdot\boxed{[e^{-3t}\cdot\dfrac{d}{dt}[-3t]]}}{[e^{-3t}]^2}$	(chain rule, exponential function rule)
$= \dfrac{[e^{-3t}]\dfrac{d}{dt}[\cos(t^2-1)] - [\cos(t^2-1)]\cdot[e^{-3t}\cdot\boxed{(-3)}]}{[e^{-3t}]^2}$	(constant multiple rule, power rule)
$= \dfrac{[e^{-3t}]\cdot\boxed{[-\sin(t^2-1)\cdot\dfrac{d}{dt}(t^2-1)]} - [\cos(t^2-1)]\cdot[e^{-3t}\cdot(-3)]}{[e^{-3t}]^2}$	(chain rule, trig function rule)
$= \dfrac{[e^{-3t}]\cdot[-\sin(t^2-1)\cdot\boxed{(2t-0)}] - [\cos(t^2-1)]\cdot[e^{-3t}\cdot(-3)]}{[e^{-3t}]^2}$	(sum rule, power rule, constant function rule)

And now I am done, because I have no more "please take the derivative of" signs left. Yay!

List of derivative shortcut rules

Keep this list for reference!

- ☐ sum rule: $\frac{d}{dx}[f(x)+g(x)] = \frac{d}{dx}[f(x)] + \frac{d}{dx}[g(x)]$
- ☐ constant multiple rule: $\frac{d}{dx}[c \cdot f(x)] = c \cdot \frac{d}{dx}[f(x)]$
- ☐ constant function rule: $\frac{d}{dx}[c] = 0$
- ☐ power rule: $\frac{d}{dx}[x^n] = nx^{n-1}$
- ☐ exponential function rule: $\frac{d}{dx}[a^x] = a^x \cdot \ln(a)$

  trigonometric function rules:

  - ◯ $\frac{d}{dx}[\sin(x)] = \cos(x)$
  - ◯ $\frac{d}{dx}[\cos(x)] = -\sin(x)$
  - ◯ $\frac{d}{dx}[\tan(x)] = \sec^2(x)$

- ☐ product rule: $\frac{d}{dx}[f(x) \cdot g(x)] = \frac{d}{dx}[f(x)] \cdot g(x) + \frac{d}{dx}[g(x)] \cdot f(x)$
- ☐ quotient rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x) \cdot \frac{d}{dx}[f(x)] - f(x) \cdot \frac{d}{dx}[g(x)]}{[g(x)]^2}$
- ☐ chain rule: $[f(g(x))]' = f'(g(x)) \cdot g'(x)$, or in other words, $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot \frac{d}{dx}[g(x)]$
- ☐ logarithmic function rule: $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$

  inverse trigonometric function rules:

  - ◯ $\frac{d}{dx}[\arctan(x)] = \frac{1}{1+x^2}$
  - ◯ $\frac{d}{dx}[\arcsin(x)] = \frac{1}{\sqrt{1-x^2}}$
  - ◯ $\frac{d}{dx}[\arccos(x)] = -\frac{1}{\sqrt{1-x^2}}$

  rules for other trigonometric functions:

  - ◯ $\frac{d}{dx}[\cot(x)] = -\csc^2(x)$
  - ◯ $\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x)$
  - ◯ $\frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$